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```

clc; clear; close all;
% System parameters
bandwidth = 10e6;
rb_size = 180e3;
% Total Resource Blocks
% 10 MHz
% 180 kHz per RB
total_RB = floor(bandwidth / rb_size);
% Users
users = 1:6;
% RB allocation (simple equal + remainder distribution)
base = floor(total_RB / length(users));
alloc = base * ones(1, length(users));
alloc(1:mod(total_RB, length(users))) = alloc(1:mod(total_RB, length(users))) +
1;
utilization = (alloc / total_RB) * 100;
disp('Total Resource Blocks:')
disp(total_RB)
disp('RB Allocation per User:')
disp(alloc)
figure
% ----- Graph 1: RB Allocation per User -----
subplot(2,1,1)
stem(users, alloc, 'filled', 'LineWidth', 2)
title('OFDMA Resource Block Allocation per User')
xlabel('User Index')
ylabel('Number of Resource Blocks')
grid on
% ----- Graph 2: RB Utilization -----
subplot(2,1,2)
plot(users, utilization, '-o', 'LineWidth', 2)
title('OFDMA Resource Block Utilization (%)')
xlabel('User Index')
ylabel('Utilization (%)')
grid on

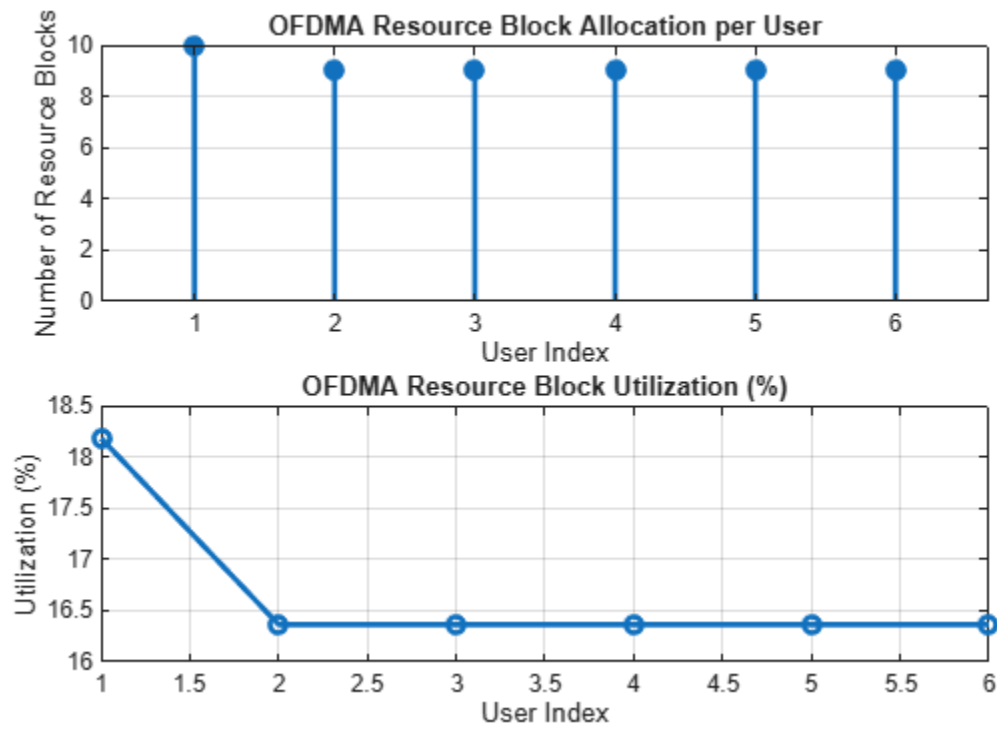
```

*Total Resource Blocks:*

55

*RB Allocation per User:*

10      9      9      9      9      9



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